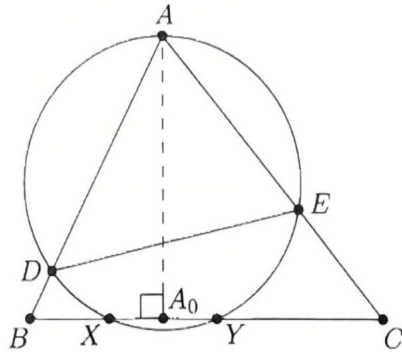


the side BC in two points X and Y . Show that the midpoint of XY is the foot of the altitude from A to BC .

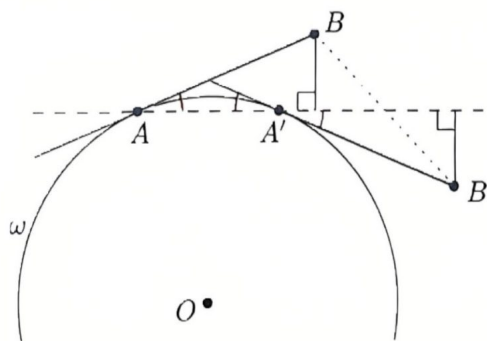
Proof. Let $A_0 \in BC$ be the foot of altitude from A . As BC and DE are antiparallel in $\angle BAC$ and AA_0 is altitude in triangle ABC , it must pass through the circumcenter of triangle ADE (H and O are friends – see Proposition 1.47).



Thus, the circumcircle of triangle ADE is symmetric in line AA_0 and since X and Y correspond in this symmetry, A_0 is the midpoint of XY .

4. [Tournament of Towns 2007] Point B lies on a line which is tangent to circle ω at point A . The line segment AB is rotated about the center of the circle by some angle to form segment $A'B'$. Prove that the line AA' bisects the segment BB' .

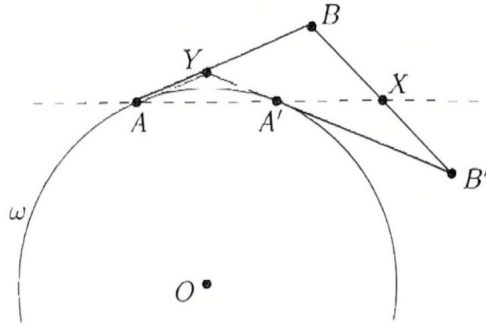
First Proof. Without loss of generality assume that AA' is horizontal. Since the segments AB and $A'B'$ form the same angle with the line AA' (Equal Tangents) and are of equal length, point B is as much “above” AA' as B' is “below” it. Then the midpoint of BB' must lie on AA' .



Second Proof. Let X be the intersection of AA' with BB' and Y the intersection of AB with $A'B'$. By Menelaus' Theorem in triangle YBB' , we have that

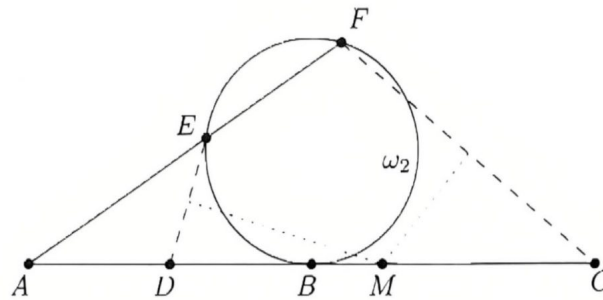
$$\frac{BX}{XB'} \cdot \frac{B'A'}{A'Y} \cdot \frac{YA}{AB} = 1.$$

But $A'Y = AY$ (Equal Tangents) and $A'B' = AB$, so we immediately get that $BX = XB'$, which is precisely what we wanted.



5. [USAMO 1998] Let ω_1 and ω_2 be concentric circles, with ω_2 in the interior of ω_1 . From a point A on ω_1 draw the tangent AB to ω_2 ($B \in \omega_2$). Let C be the second point of intersection of AB and ω_1 , and let D be the midpoint of AB . A line passing through A intersects ω_2 at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB . Find the ratio AM/MC .

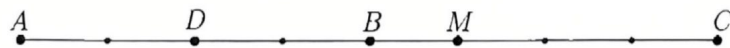
Solution. We place AB horizontally and remove the circle ω_1 entirely keeping in mind that B is the midpoint of AC , which follows from symmetry. With what is left in the picture now, Power of a Point is a must.



It yields

$$AE \cdot AF = AB^2 = \frac{AB}{2} \cdot 2AB = AD \cdot AC$$

implying that $EFCD$ is cyclic and thus M as the intersection of two perpendicular bisectors in $EFCD$ is inevitably its circumcenter. In particular, it is the midpoint of CD . Now we calculate the desired ratio easily as



$$AM = AD + \frac{1}{2}CD = \frac{1}{4}AC + \frac{3}{8}AC = \frac{5}{8}AC,$$

then $MC = \frac{3}{8}AC$ and the answer is $\frac{5}{3}$.

6. [Titu Andreescu] Let M be a point inside triangle ABC such that

$$AM \cdot BC + BM \cdot AC + CM \cdot AB = 4[ABC].$$

Show that M is the orthocenter of triangle ABC .

Proof. We aim to relate the products on the left-hand side to some areas.

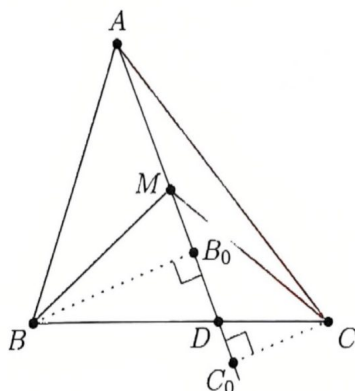
Extend AM to meet BC at D and denote by B_0, C_0 the feet of perpendiculars dropped onto AM from B, C , respectively. Since perpendicular is the shortest distance from a point to a line,

$$BC = BD + DC \geq BB_0 + CC_0$$

and

$$AM \cdot BC \geq AM \cdot BB_0 + AM \cdot CC_0 = 2 \cdot ([AMB] + [CMA]),$$

with equality if and only if $AM \perp BC$.



Likewise we obtain

$$BM \cdot AC \geq 2 \cdot ([BMC] + [AMB]),$$

$$CM \cdot AB \geq 2 \cdot ([CMA] + [BMC]).$$

Summing these three inequalities we get

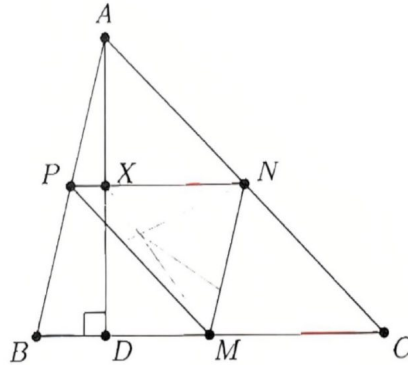
$$\begin{aligned} AM \cdot BC + BM \cdot AC + CM \cdot AB &\geq 4([AMB] + [BMC] + [CMA]) \\ &= 4[ABC]. \end{aligned}$$

Hence the equality in all three partial inequalities has to occur, implying that M is the orthocenter of triangle ABC .

7. Let ABC be a triangle. Prove that lines joining midpoints of the sides with midpoints of the corresponding altitudes pass through a single point.

Proof. Denote the midpoints of the sides BC , CA , AB by M , N , P , respectively, the feet of corresponding altitudes by D , E , F , and their midpoints by X , Y , Z , respectively.

Since X is the midpoint of AD , it lies on the midline NP of triangle ABC . Hence it is convenient to view the situation with respect to triangle MNP .



In order to establish the concurrence, by Ceva's Theorem it suffices to prove

$$\frac{NX}{XP} \cdot \frac{PY}{YM} \cdot \frac{MZ}{ZN} = 1.$$

Next we get rid of all the midpoints. As NX and XP are midlines in triangles CAD and DAB , we have $NX = \frac{1}{2}CD$ and $XP = \frac{1}{2}DB$, and thus the first ratio on the left-hand side equals CD/DB . By similar arguments we rewrite the other two ratios and learn

$$\frac{NX}{XP} \cdot \frac{PY}{YM} \cdot \frac{MZ}{ZN} = \frac{CD}{DB} \cdot \frac{AE}{EC} \cdot \frac{BF}{FA}$$

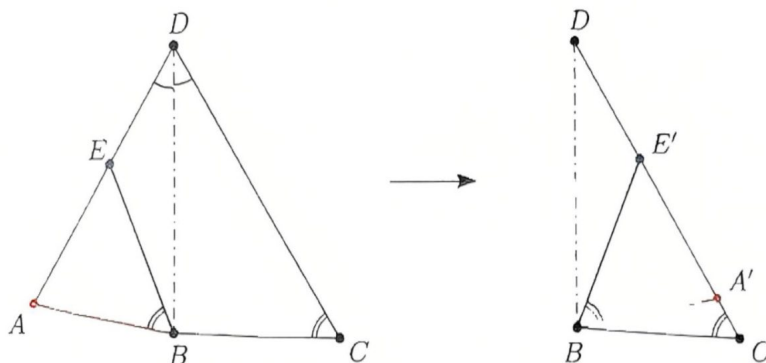
But the latter is equal to one by another Ceva's Theorem because the segments AD , BE , CF actually are concurrent cevians in triangle ABC (they concur at its orthocenter) so the proof is complete.

8. [Baltic Way 2011] Let $ABCD$ be a convex quadrilateral such that $\angle ADB = \angle BDC$. Suppose that a point E on the side AD satisfies the equality

$$AE \cdot ED + BE^2 = CD \cdot AE.$$

Show that $\angle EBA = \angle DCB$.

Proof. We draw DB vertically to be more aware of the symmetry. Observe that most distances in the given metric condition take place either on line DC or on the symmetric line DA . It should strike us that we could restate the problem by putting all distances just on one of the symmetric lines.



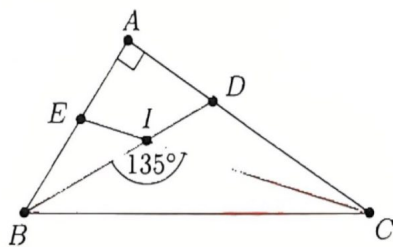
Indeed, let A' and E' be the reflections of A and E over BD . It's not a surprise that the condition simplifies:

$$BE'^2 = BE^2 = CD \cdot AE - AE \cdot ED = A'E'(CD - E'D) = A'E' \cdot CE'.$$

By Power of a Point, $E'B$ is tangent to the circumcircle of triangle $BA'C$ and so $\angle A'BE' = \angle DCB$ (see Proposition 1.34). The conclusion follows by symmetry.

9. [USAMO 2010] Let ABC be a triangle with $\angle A = 90^\circ$. Denote its incenter by I and let $D = BI \cap AC$ and $E = CI \cap AB$. Determine whether or not it is possible for segments AB , AC , BI , ID , CI , IE to all have integer lengths.

Proof. Having in mind basic angles in a triangle, we aim to make use of $\angle BIC = 135^\circ$ (see Proposition 1.11). The trick is to express $\cos \angle BIC$ from the Law of Cosines in triangle BIC .



With the help of the Pythagorean Theorem in triangle ABC , we obtain

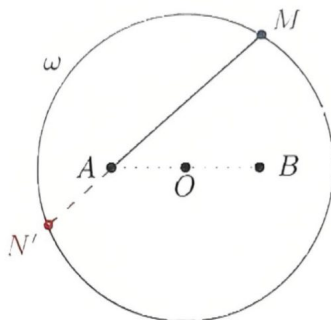
$$-\frac{\sqrt{2}}{2} = \cos 135^\circ = \frac{BI^2 + CI^2 - BC^2}{2BI \cdot CI} = \frac{BI^2 + CI^2 - AB^2 - AC^2}{2BI \cdot CI}.$$

We see that the left-hand side is irrational, thus it is impossible for all the distances AB , AC , BI , and CI to have integer lengths.

10. Let A and B be two fixed points inside of the fixed circle ω symmetric with respect to its center O . If points M and N vary on ω in the same

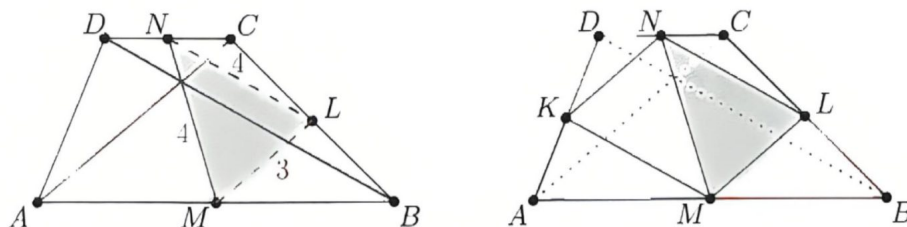
half-plane with respect to AB , so that $AM \parallel BN$, prove that $AM \cdot BN$ is constant.

Proof. Extend MA to meet ω for the second time at N' . As A and B are symmetric about O and $AM \parallel BN$, points N and N' are also symmetric about O and $AN' = BN$. Thus the product $AM \cdot BN$ equals $AM \cdot AN'$ which is simply (negative) power of A with respect to ω , a constant.



11. In a trapezoid $ABCD$, the segment connecting the midpoints M, N of the bases AB, CD , respectively, has length 4, and the diagonals have lengths $AC = 6$ and $BD = 8$. Find the area of the trapezoid.

First Solution. Let L be the midpoint of BC . Then $ML = \frac{1}{2}AC = 3$ and $LN = \frac{1}{2}BD = 4$ so Heron's formula yields $[MLN] = \frac{3}{4}\sqrt{55}$. Next we relate $[MLN]$ to $[ABCD]$.



Let K be the midpoint of AD . As $KMLN$ is a parallelogram (see Introductory Problem 9), $[KMLN] = 2 \cdot [MLN]$. Regarding the areas of the remaining four small triangles, since ML and KN are midlines, we have

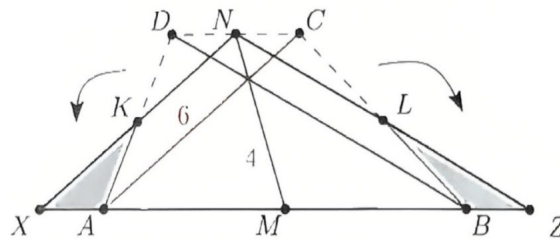
$$[MBL] + [KDN] = \frac{1}{4}[ABC] + \frac{1}{4}[ADC] = \frac{1}{4}[ABCD]$$

and likewise $[KAM] + [NCL] = \frac{1}{4}[ABCD]$. Thus the parallelogram $KMLN$ occupies $1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$ of the area of $ABCD$ and finally

$$[ABCD] = 2 \cdot [KMLN] = 4 \cdot [MLN] = 3\sqrt{55}.$$

Second Solution. As in the first solution, let K, L be the midpoints of the sides AD, BC , respectively.

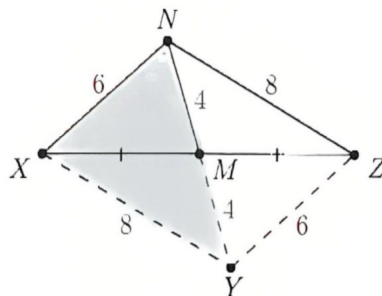
We cut off the triangle LCN and flip it about L to obtain triangle LBZ (this is possible as $LB = LC$). Since $ABCD$ is a trapezoid, Z lies on the line AB , and we have $MZ = \frac{1}{2}AB + \frac{1}{2}CD$ and $NZ = 2 \cdot NL = DB = 8$.



Doing the same for triangle KDN (i.e. flipping triangle KDN about K into triangle KAX) we realize that it suffices to determine the area of a triangle NXZ given the lengths of two of its sides NZ , NX , and the length of its median NM .

Let Y be the point such that $NXYZ$ is a parallelogram. Then $[NXZ] = \frac{1}{2}[NXYZ] = [NXY]$ and the side lengths of triangle NXY are known as $NX = 6$, $XY = NZ = 8$, and $NY = 2 \cdot NM = 8$. Hence we conclude using Heron's formula:

$$[ABCD] = [NXZ] = [NXY] = \sqrt{11 \cdot 5 \cdot 3 \cdot 3} = 3\sqrt{55}.$$



12. In triangle ABC , let AP , BQ , CR be concurrent cevians. Let the circumcircle of triangle PQR intersect the sides BC , CA , AB for the second time at X , Y , Z , respectively. Prove that AX , BY , CZ are concurrent.

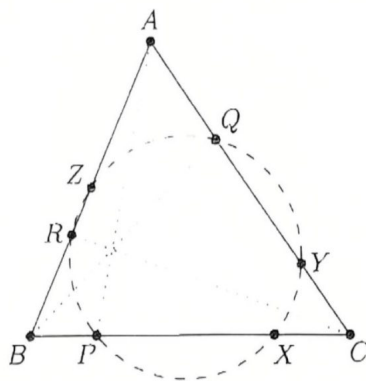
Proof. We start with Ceva's Theorem for concurrent cevians AP , BQ , CR and we aim to apply another Ceva's Theorem for AX , BY , CZ . We expect to handle the circle by means of Power of a Point.

We have

$$AZ \cdot AR = AQ \cdot AY \quad \text{and thus} \quad \frac{AR}{AQ} = \frac{AY}{AZ}.$$

Likewise we obtain

$$\frac{BP}{BR} = \frac{BZ}{BX}, \quad \frac{CQ}{CP} = \frac{CX}{CY}.$$



Now we choose to write the aforementioned Ceva's Theorem for AP , BQ , CR in the form

$$\frac{AR}{AQ} \cdot \frac{BP}{BR} \cdot \frac{CQ}{CP} = 1.$$

Substituting the derived equalities yields

$$\frac{AY}{AZ} \cdot \frac{BZ}{BX} \cdot \frac{CX}{CY} = 1,$$

which is equivalent to the desired metric condition from Ceva's Theorem for AX , BY , and CZ . We may conclude.

Remark. This is a particular case of a celebrated theorem by Carnot that if P , X lie on BC , Q , Y on CA , and R , Z on AB , then the points P , Q , R , X , Y , Z all lie on the same conic if and only if

$$\frac{XB}{XC} \cdot \frac{PB}{PC} \cdot \frac{YC}{YA} \cdot \frac{QC}{QA} \cdot \frac{ZA}{ZB} \cdot \frac{RA}{RB} = 1.$$

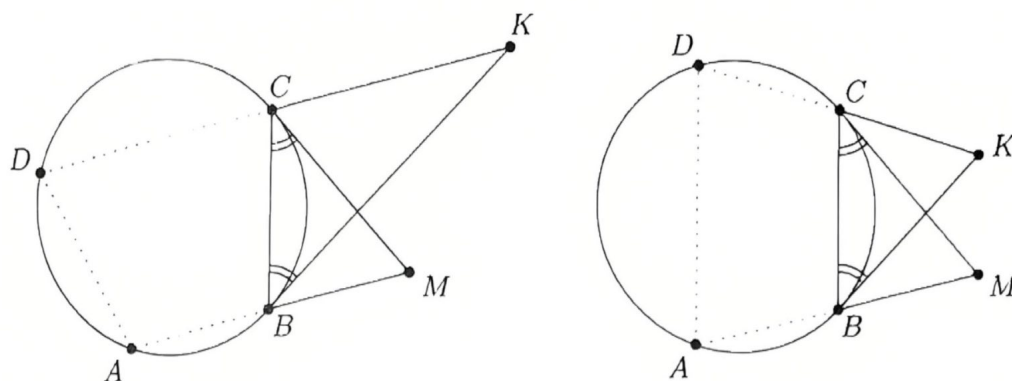
13. [All-Russian Olympiad 2002] A quadrilateral $ABCD$ is inscribed in a circle ω . The tangent to ω at B intersects the ray DC at K , and the tangent to ω at C intersects the ray AB at M . Prove that if $BM = BA$ and $CK = CD$, then $ABCD$ is a trapezoid.

First Proof. The midpoints suggest we should try to use concyclicity of $ABCD$ in terms of ratios.

By Power of a Point we learn $KB^2 = KC \cdot KD = 2KC^2$ and similarly $MC^2 = MB \cdot MA = 2MB^2$. Thus the ratios KB/KC and MC/MB are equal. Moreover, both angles $\angle BCM = \angle KBC$ correspond to the same arc BC and thus are also equal.

Now it is time to observe that triangles KCB and MBC have a lot in common. We exploit this by the Law of Sines:

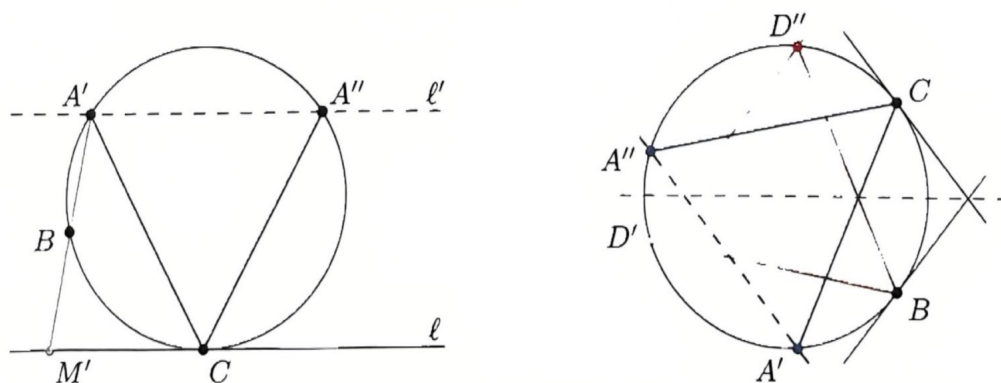
$$\frac{KB}{KC} = \frac{\sin \angle KCB}{\sin \angle KBC}, \quad \frac{MC}{MB} = \frac{\sin \angle MBC}{\sin \angle BCM},$$



from which we learn $\sin \angle KCB = \sin \angle MBC$. We distinguish two cases. If $\angle KCB = \angle MBC$, then $BC \parallel DA$ and if $\angle KCB = 180^\circ - \angle MBC$, then $AB \parallel CD$. Either way, $ABCD$ is a trapezoid and we may conclude.

Second Proof. We approach the problem as a ruler-compass construction. Starting only with ω and points B and C , we are looking for suitable choices of A and D . Since B is to be the midpoint of AM , the point A must lie on line ℓ' which is symmetric in point B with the tangent ℓ to ω at C .

Let the two intersections of ℓ' and ω be A' and A'' , the only possible positions of point A . Since $\ell \parallel \ell'$, point C is the midpoint of arc $A'A''$ (containing B) and hence $A'C = A''C$.

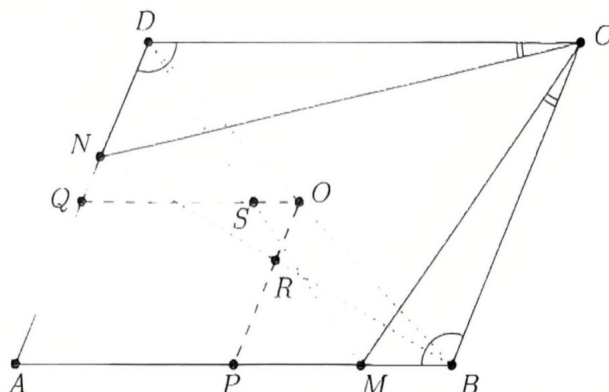


We have shown that the diagonal AC has equal length for both positions of A . The same holds for the diagonal BD . Moreover, these lengths are equal as we can see from line symmetry in the axis of BC . Thus, $ABCD$ has equal diagonals so it is indeed a trapezoid (if in doubt, see Example 1.7).

14. Let $ABCD$ be a parallelogram and M, N points on its sides AB, AD such that $\angle MCB = \angle DCN$. Let P, Q, R , and S be the midpoints of the segments AB, AD, NB , and MD , respectively. Show that P, Q, R , and S are concyclic.

Proof. Our first observation is that PR is the midline in triangle ABN

and QS is the midline in triangle ADM . These midlines are in fact the midlines of parallelogram $ABCD$ and intersect at its center O .



We will show concyclicity by means of Power of a Point. We want to prove

$$OR \cdot OP = OS \cdot OQ.$$

As OR and OP are midlines in triangles DBN and DBA , we see that $OR = \frac{1}{2}DN$ and $OP = \frac{1}{2}DA$. Similarly, we derive $OS = \frac{1}{2}BM$ and $OQ = \frac{1}{2}BA$. Now it suffices to prove

$$DN \cdot DA = BM \cdot BA \quad \text{or} \quad \frac{DN}{DC} = \frac{BM}{BC},$$

but the last relation follows from the similarity of triangles MBC and NDC (AA). We are done.

15. [Tournament of Towns 2008] Diagonals of non-isosceles trapezoid $ABCD$ intersect at P . Let A_1 be the second intersection of the circumcircle of triangle BCD and AP . Points B_1, C_1, D_1 are defined in a similar way. Prove that $A_1B_1C_1D_1$ is also a trapezoid.

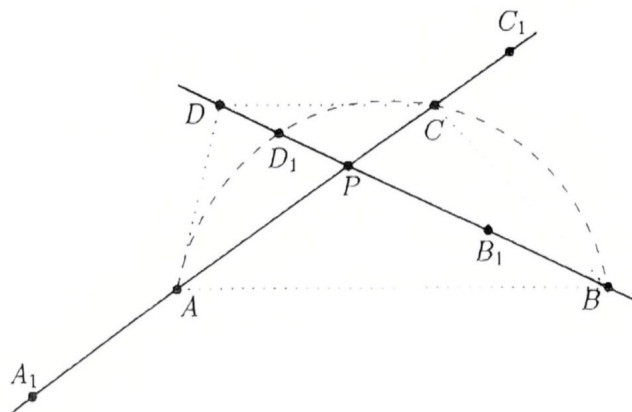
Proof. It suffices to consider the case when $AB \parallel CD$. Denote the lengths of the segments PA, PB, PC, PD by a, b, c, d , respectively. We intend to locate the positions of the points A_1, B_1, C_1 , and D_1 on the diagonals AC and BD by Power of a Point and then to use similarity to work with the trapezoids.

Taking power of P with respect to the four circles yields

$$PA_1 = \frac{bd}{c}, \quad PB_1 = \frac{ac}{d}, \quad PC_1 = \frac{bd}{a}, \quad \text{and} \quad PD_1 = \frac{ac}{b}.$$

From $AB \parallel CD$ we infer $\triangle ABP \sim \triangle CDP$ and $b/a = d/c$. Hence equations

$$\frac{PA_1}{PB_1} = \frac{bd}{c} \cdot \frac{d}{ac} = \frac{bd}{ac} \cdot \frac{d}{c} \quad \text{and} \quad \frac{PC_1}{PD_1} = \frac{bd}{a} \cdot \frac{b}{ac} = \frac{bd}{ac} \cdot \frac{b}{a}$$

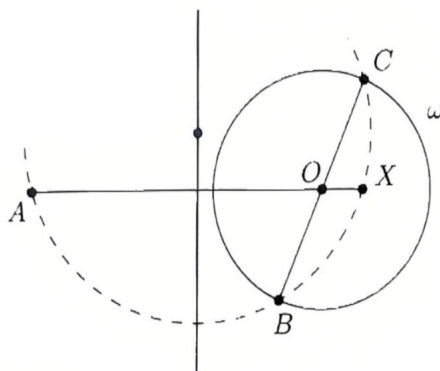


imply that $PA_1/PB_1 = PC_1/PD_1$. As a consequence, the triangles A_1B_1P and C_1D_1P are similar (SAS) and $A_1B_1C_1D_1$ is indeed a trapezoid.

16. [Czech and Slovak 2006] Let ω be a circle with center O and radius r and A a point different from O . Find the locus of circumcenters of the triangles ABC for which BC is a diameter of ω .

Solution. Here the key is the following observation. The point O has constant power with respect to all circumcircles of ABC , namely $-OB \cdot OC = -r^2$. Then the second intersection X of line AO with the circumcircle of ABC is fixed, since from

$$OA \cdot OX = OB \cdot OC$$



we infer $OX = r^2/OA$. Therefore all the circles pass through two fixed points, hence their centers lie on a fixed line (namely the perpendicular bisector of AX). In order to show that any point of this line belongs to the desired locus, it suffices to note that any circle through A and X intersects ω at two diametrically opposite points, which can be seen from reversing the performed calculations.

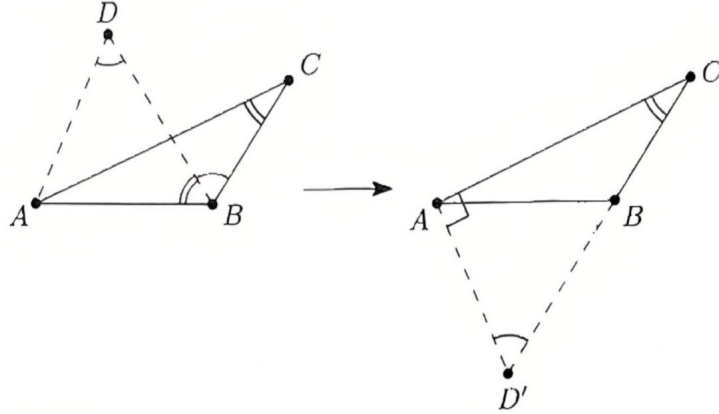
17. Let $ABCD$ be quadrilateral such that

$$\angle ADB + \angle ACB = 90^\circ \quad \text{and} \quad \angle DBC + 2\angle DBA = 180^\circ.$$

Show that

$$(DB + BC)^2 = AD^2 + AC^2.$$

Proof. The strange angular conditions and the form of the conclusion resembling the Pythagorean Theorem call for some transformation. First, we erase the segment CD . Then we look at the figure as if it was folded along the line AB . Unfolding does the trick!



Indeed, if D' is the mirror image of D over AB , then

$$\angle CBD' = \angle CBD + 2 \cdot \angle DBA = 180^\circ,$$

and the points C , B , and D' are collinear. Also, in triangle $AD'C$ we have $\angle D'AC = 180^\circ - \angle ACB - \angle BD'A = 90^\circ$, hence it is right. The conclusion now follows from the Pythagorean Theorem (and symmetry).

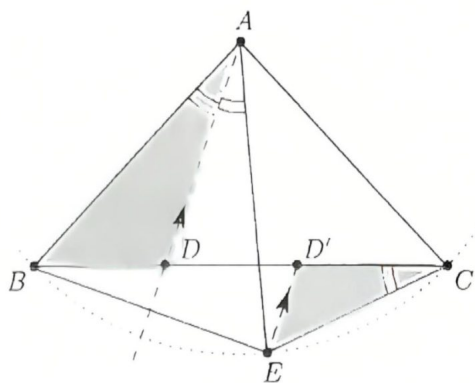
18. [Poland 2008] We are given a triangle ABC such that $AB = AC$. There is a point D lying on the segment BC , such that $BD < DC$. Point E is symmetrical to B with respect to AD . Prove that

$$\frac{AB}{AD} = \frac{CE}{CD - BD}.$$

First Proof. Let D' be the point on segment BC with $D'C = BD$. Then $DD' = CD - D'C = CD - BD$. By symmetry we have $\angle BAD = \angle D'AC$ and if we also use the symmetry about AD , we learn $\angle DAD' = \angle BAC - 2\angle BAD = \angle EAC$.

Thus triangles $D'AD$ and EAC being isosceles with the same vertex angle are similar. Hence

$$\frac{AB}{AD} = \frac{AC}{AD'} = \frac{CE}{DD'} = \frac{CE}{CD - BD}$$

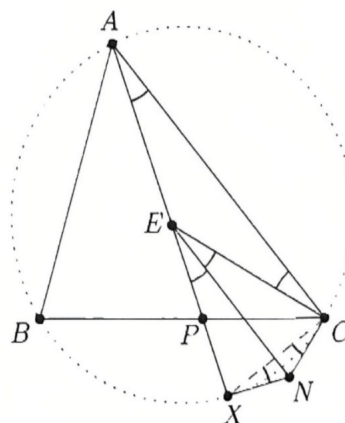
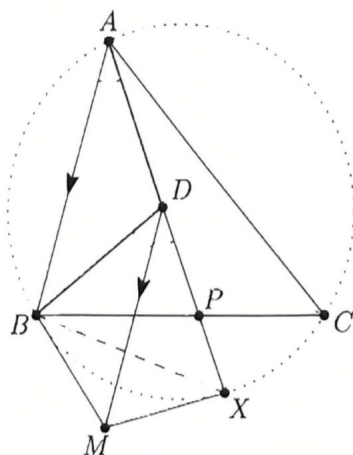


$BD'E$, thus $AD \parallel D'E$. Then $\angle ADB = \angle ED'C$, so now we indeed have $\triangle ABD \sim \triangle CED'$ (AA), which yields the desired

$$\frac{AB}{AD} = \frac{CE}{CD'} = \frac{CE}{CD - BD}.$$

19. Let P be a point on the side BC of triangle ABC . Perpendicular bisectors of the sides AB and AC meet the segment AP at points D and E , respectively. The line parallel to AB passing through D intersects the tangent to the circumcircle ω of triangle ABC through B at point M . Similarly, the line parallel to AC passing through E intersects the tangent to ω through C at point N . Prove that MN is tangent to ω .

Proof. Everything seems to be arranged for angle-chasing. As usual, we do not draw the perpendicular bisectors and work with isosceles triangles BDA and CEA instead. First, we focus on the left part of the picture. From the isosceles triangle BDA we have $\angle BAD = \angle DBA$, then as $DM \parallel AB$, also $\angle BDM = \angle DBA$. Next, $\angle BDP$ is external angle in triangle BDA , thus $\angle BDP = 2\angle BAD$, and so $\angle MDP = \angle BDP - \angle BDM = \angle BAD$.



Now comes the vital step. Our common angle chasing tools can only prove that a line is tangent at a certain point. But here, we have no

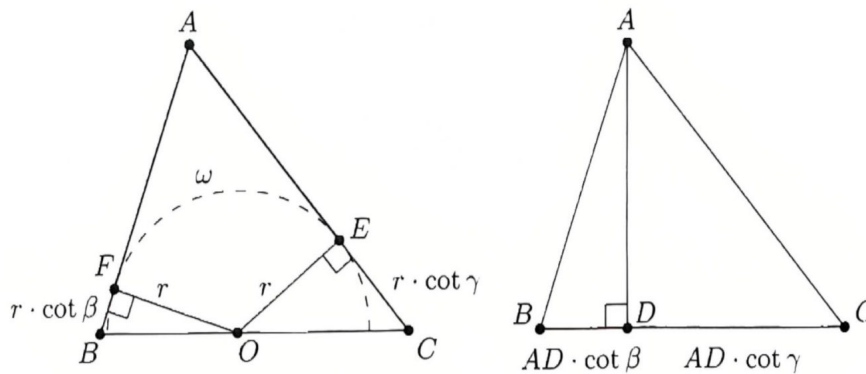
information about the point of contact. We must guess where it is. With good intuition or some skill in mental angle-chasing, we choose the second intersection X of AP and ω . Then since BM is tangent to ω we have $\angle MBX = \angle BAX$ (see Proposition 1.34), and so $\angle MBX = \angle MDX$ therefore $BMXD$ is cyclic. Finally, we deduce that $\angle BXM = \angle BDM = \angle BAP$ which implies that MX is tangent to ω .

Analogously, we prove that NX is tangent to ω , therefore MN is a tangent to ω at X .

20. In an acute triangle ABC a semicircle ω centered on the side BC is tangent to the sides AB and AC at points F and E , respectively. If X is the intersection of BE and CF , show that $AX \perp BC$.

Proof. If we denote by D the foot of the A -altitude, we may change the perspective and switch to proving that cevians AD , BE , and CF concur. By Ceva's Theorem we need to verify

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$



which since $AE = AF$ (Equal Tangents) reduces to

$$\frac{BD}{DC} = \frac{FB}{CE},$$

Now we denote the center of ω by O and its radius by r . Right triangles OFB and OEC then give

$$\frac{FB}{CE} = \frac{r \cdot \cot \beta}{r \cdot \cot \gamma} = \frac{\cot \beta}{\cot \gamma}$$

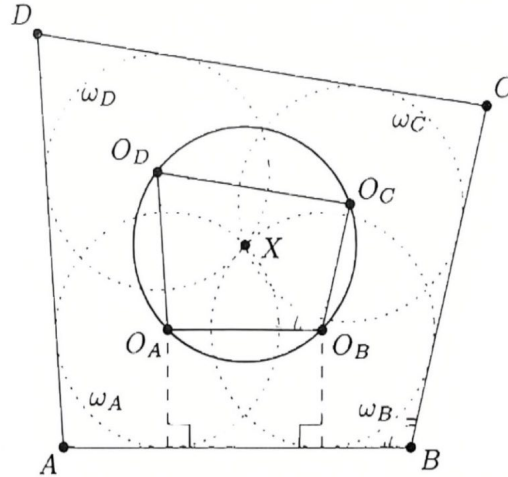
and similarly right triangles BDA and DCA give

$$\frac{BD}{DC} = \frac{AD \cdot \cot \beta}{AD \cdot \cot \gamma} = \frac{\cot \beta}{\cot \gamma}$$

which ends the proof.

21. Let $ABCD$ be a convex quadrilateral and X a point in its interior. Denote by ω_A the circle tangent to the sides AB and AD and passing through X . Define circles ω_B , ω_C , and ω_D similarly. Given that all these circles have equal radii, show that $ABCD$ is cyclic.

Proof. We denote the centers of ω_A , ω_B , ω_C , and ω_D , by O_A , O_B , O_C , and O_D , respectively. Now since the circles have equal radii and all pass through X , we have $O_AX = O_BX = O_CX = O_DX$, thus points O_A , O_B , O_C , and O_D lie on a circle with center X .



It remains to observe that $O_AO_B \parallel AB$, since points O_A , O_B have equal distance from the line AB . Similarly, we have $O_BO_C \parallel BC$, $O_CO_D \parallel CD$, and $O_DO_A \parallel DA$. Now it can be easily seen that

$$\angle CBA = \angle O_CO_BO_A = 180^\circ - \angle O_AO_DO_C = 180^\circ - \angle ADC,$$

hence $ABCD$ is cyclic as desired.

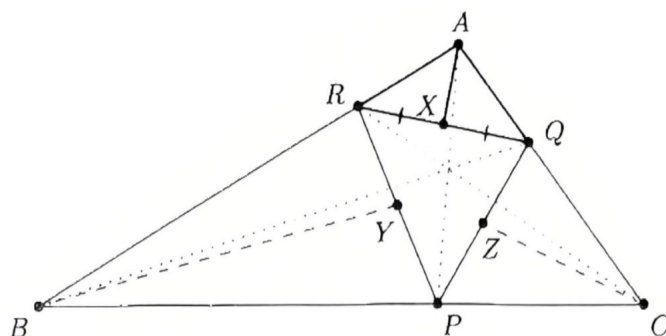
22. [Mathematical Reflections, Ivan Borsenco] In triangle ABC , let AP , BQ , CR be concurrent cevians. Denote by X , Y , Z the midpoints of segments QR , RP , PQ , respectively. Prove that the lines AX , BY , CZ are concurrent.

Proof. We intend to use the trigonometric form of Ceva's Theorem. From the Ratio Lemma (see Proposition 1.18) for triangle ARQ we learn

$$\frac{AR \sin \angle RAX}{QA \sin \angle XAQ} = \frac{RX}{XQ} = 1,$$

hence

$$\frac{\sin \angle RAX}{\sin \angle XAQ} = \frac{QA}{AR}.$$



In the same vein, we find two analogous relations and we may write

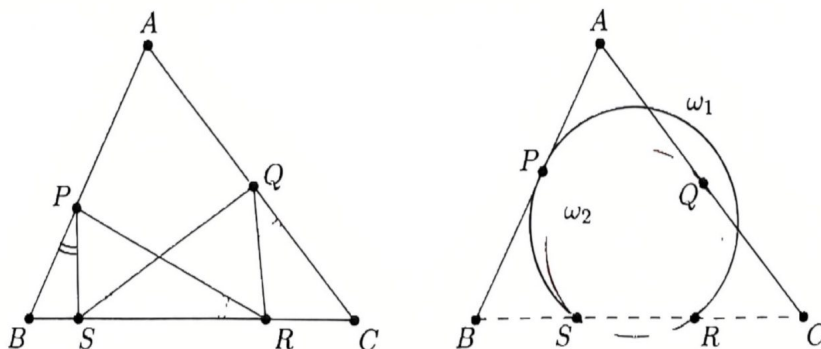
$$\frac{\sin \angle RAX}{\sin \angle XAQ} \cdot \frac{\sin \angle PBX}{\sin \angle YBR} \cdot \frac{\sin \angle QCZ}{\sin \angle ZCP} = \frac{QA}{AR} \cdot \frac{RB}{BP} \cdot \frac{PC}{CQ},$$

where the right-hand side is equal to 1 by Ceva's Theorem applied to concurrent cevians AP , BQ , and CR . Thus also the left-hand side equals 1 and we are done by the trigonometric form of Ceva's Theorem.

Remark. The conclusion of the theorem still holds if we take any triplet of points X , Y , Z on the sides QR , RP , PQ of triangle PQR for which the lines PX , QY , RZ concur. The result is called the *Cevian Nest Theorem* and is proved with the same technique as this problem.

23. [USAJMO 2012] Given a triangle ABC , let P and Q be points on segments AB and AC , respectively, such that $AP = AQ$. Let S and R be distinct points on segment BC such that S lies between B and R , $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that P , Q , R , S are concyclic.

Proof. We interpret the angle relations as tangency. From $\angle BPS = \angle PRS$ we infer that BP is tangent to the circumcircle ω_1 of triangle SRP and $\angle CQR = \angle QSR$ implies CQ is tangent to the circumcircle ω_2 of triangle RQS (see Proposition 1.34). We are asked to prove that circles ω_1 and ω_2 coincide. Let us assume they are distinct.



Then their radical axis is the line SR and at the same time

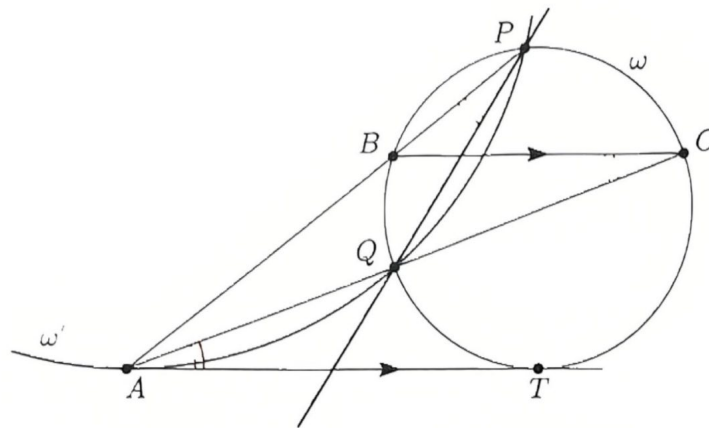
$$p(A, \omega_1) = AP^2 = AQ^2 = p(A, \omega_2).$$

Since $A \notin BC$, we reached a contradiction.

24. Segment AT is tangent to circle ω at T . A line parallel to AT intersects ω at B, C (with $AB < AC$). Lines AB, AC intersect ω for the second time at P, Q . Prove that line PQ bisects segment AT .

Proof. Suppose that points A, B, P , and A, Q, C are collinear in this order (the other cases are analogous). Since $AT \parallel BC$ and $BQCP$ is cyclic, we have

$$\angle TAC = \angle BCA \equiv \angle BCQ = \angle BPQ.$$



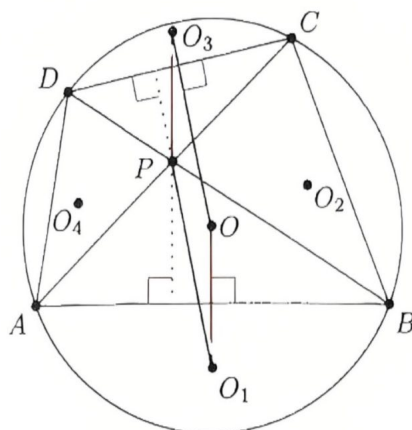
This implies that AT is tangent to the circumcircle ω' of triangle APQ (see Proposition 1.34). And now we are done because the radical axis PQ of ω and ω' bisects their common tangent AT (see Proposition 1.41).

25. [China 1990] Diagonals AC and BD of a cyclic quadrilateral $ABCD$ meet at P . Let the circumcenters of $ABCD$, ABP , BCP , CDP , and DAP be O, O_1, O_2, O_3 , and O_4 , respectively. Prove that OP, O_1O_3 , and O_2O_4 are concurrent.

Proof. We claim that PO_1OO_3 is a parallelogram. First, since AB and CD are antiparallel in $\angle APB$ and since “ H and O are friends” (see Proposition 1.47), the line O_1P is also the altitude in triangle CPD , thus $O_1P \perp CD$.

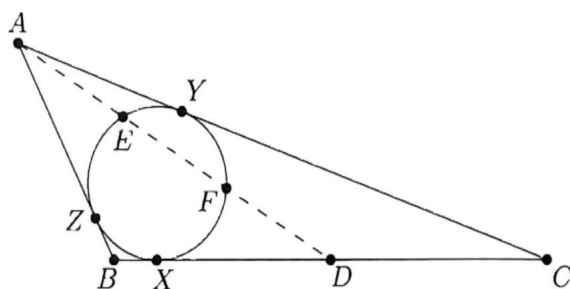
But also $OO_3 \perp CD$, since both circumcenters lie on the perpendicular bisector of CD . Hence $OO_3 \parallel O_1P$. Similarly, we prove that $OO_1 \parallel O_3P$ and so PO_1OO_3 is indeed a parallelogram.

For analogous reasons OO_2PO_4 is also a parallelogram and since diagonals of a parallelogram bisect each other, the lines OP, O_1O_3 , and O_2O_4 are concurrent at the midpoint of OP .



26. [AIME 2005] Triangle ABC has $BC = 20$. The incircle of the triangle evenly trisects the median AD at points E and F . Find the area of the triangle.

First Solution. As an A -isosceles triangle certainly does not have the property, we may assume $b > c$. Also let the incircle touch the sides BC , CA , AB at points X , Y , and Z , respectively.



We will calculate the triangle side lengths with the help of Power of a Point, but first we recall basic distances in a triangle (see Proposition 1.15(a)) and find that

$$AZ = \frac{b+c-a}{2}, \quad DX = \frac{a}{2} - BX = \frac{a}{2} - \frac{a+c-b}{2} = \frac{b-c}{2}.$$

Taking powers of A and D with respect to the incircle we learn

$$AE \cdot AF = AZ^2 \quad \text{or} \quad \left(\frac{1}{3}AD\right) \left(\frac{2}{3}AD\right) = \left(\frac{b+c-a}{2}\right)^2,$$

and likewise

$$DF \cdot DE = DX^2 \quad \text{or} \quad \left(\frac{1}{3}AD\right) \left(\frac{2}{3}AD\right) = \left(\frac{b-c}{2}\right)^2.$$

As the left-hand sides are equal, we may compare the right-hand sides and find that $2c = a$ i.e. $c = 10$. Next, we use the median formula (see

Corollary 1.24) for AD and plugging it in we set a quadratic equation for b :

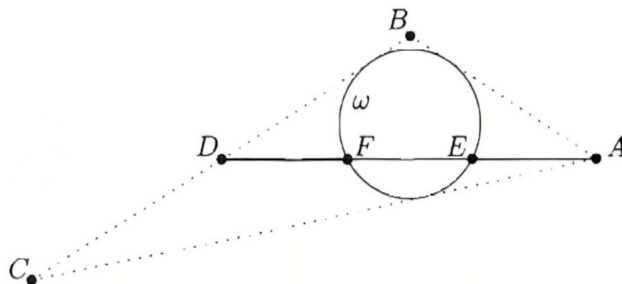
$$\frac{2}{9} \cdot \left(\frac{b^2 + 10^2}{2} - \frac{20^2}{4} \right) = \left(\frac{b - 10}{2} \right)^2.$$

Simplifying we get $b^2 - 36b + 260 = 0$ with solution $b = 26$ (and also $b = 10$ but this does not satisfy $b > c$).

Finally, we calculate the area K from Heron's formula

$$K = \sqrt{8 \cdot 2 \cdot 18 \cdot 28} = 24\sqrt{14}.$$

Second Solution. First, let us draw the median AD and the incircle ω only. Since the median is trisected evenly, the whole figure is symmetric about the perpendicular bisector of AD . Hence the tangents to ω issued from A and D , which touch ω in the same half-plane determined by AD , form an isosceles triangle.



It remains to observe that the intersection of these tangents coincides with one of the vertices B, C of the triangle ABC . If we without loss of generality assume that it is B , we obtain $AB = BD = \frac{1}{2}BC$. We continue as in the first solution.

27. [IMO 1996 shortlist, Titu Andreescu] Let P be a point inside an equilateral triangle ABC . Let the lines AP, BP, CP meet the sides BC, CA, AB at the points A_1, B_1, C_1 , respectively. Prove that

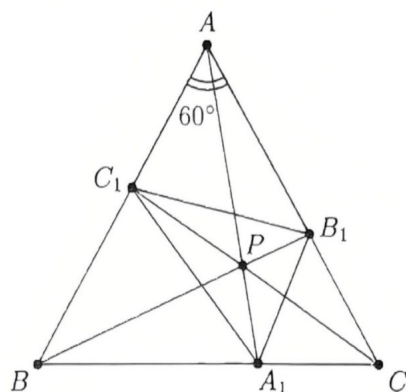
$$A_1B_1 \cdot B_1C_1 \cdot C_1A_1 \geq A_1B \cdot B_1C \cdot C_1A.$$

Proof. Ceva's Theorem for concurrent lines AA_1, BB_1, CC_1 gives

$$A_1B \cdot B_1C \cdot C_1A = A_1C \cdot B_1A \cdot C_1B$$

which allows us to be proving a more symmetrical statement. Indeed, plugging it in the square of the sought-after inequality shows that it suffices to prove

$$(A_1B_1 \cdot B_1C_1 \cdot C_1A_1)^2 \geq A_1B \cdot A_1C \cdot B_1A \cdot B_1C \cdot C_1A \cdot C_1B.$$



Applying the Law of Cosines to triangle AB_1C_1 we obtain

$$B_1C_1^2 = C_1A^2 + B_1A^2 - C_1A \cdot B_1A$$

and using the inequality $x^2 + y^2 - xy \geq xy$, which holds for all real numbers x, y , we get

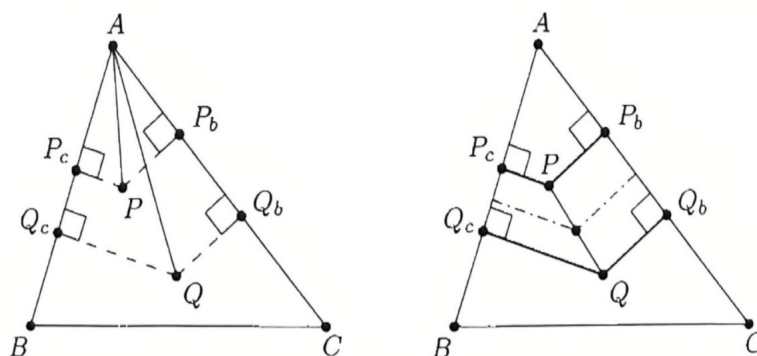
$$B_1C_1^2 \geq C_1A \cdot B_1A.$$

After obtaining two analogous inequalities and multiplying all three, we get the result. Equality holds if and only if $CA_1 = CB_1$, $AB_1 = AC_1$, and $BC_1 = BA_1$, which in turn holds if and only if P is the center of the triangle ABC .

28. Let P and Q be isogonal conjugates¹ with respect to the triangle ABC . Show that the six feet of perpendiculars from P and Q to the sides of triangle ABC lie on one circle.

Denote the feet from P to BC, CA, AB by P_a, P_b , and P_c , respectively, and define Q_a, Q_b, Q_c analogously.

First Proof. Let $\angle BAP = \angle QAC = \varphi$ and $\angle BAQ = \angle PAC = \psi$. First we use Power of a Point to show that P_c, Q_c, P_b, Q_b are concyclic.



From right triangles APP_c and AQQ_c we have

$$AP_c \cdot AQ_c = (AP \cos \varphi) \cdot (AQ \cos \psi)$$

¹For explanation see Theorem 1.46.

and similarly from APP_b and AQQ_b we obtain

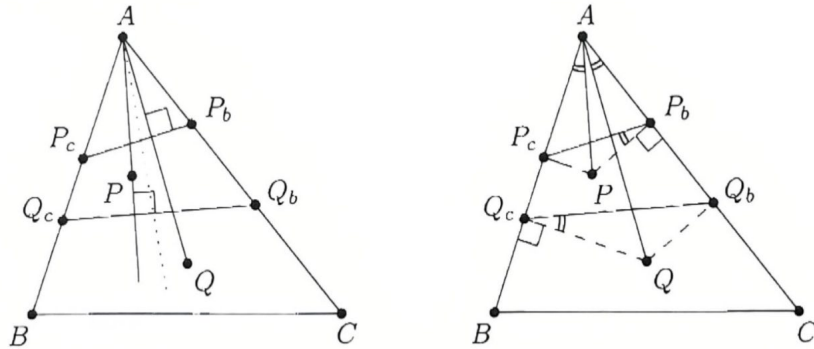
$$AP_b \cdot AQ_b = (AP \cos \psi) \cdot (AQ \cos \varphi).$$

Since the products on the right-hand sides are equal, points P_c, Q_c, P_b, Q_b lie on a single circle.

Moreover, the center of this circle is the intersection of perpendicular bisectors of P_cQ_c and P_bQ_b , which are both midlines in trapezoids P_cQ_cQP and P_bQ_bQP and thus meet at the midpoint of PQ .

Analogously, we can show that P_a, Q_a, P_c, Q_c lie on a circle centered at the midpoint of PQ . The two circles share a center and two points on their perimeters and thus coincide. All six feet then lie on one circle.

Second Proof. As lines AP and AQ are isogonal in $\angle A$ and AP (being the diameter of the circumcircle of AP_cPP_b) passes through the circumcenter of AP_cP_b , it follows (H and O are friends – see Proposition 1.47) that $AQ \perp P_bP_c$. Similarly, we can show that $AP \perp Q_bQ_c$. Finally, since AP and AQ are antiparallel in $\angle A$, the perpendicular lines, namely P_bP_c and Q_bQ_c , are also antiparallel. Hence P_c, Q_c, P_b, Q_b lie on a single circle (call it ω_a).



Now we could join the first proof but we proceed differently. As above we deduce that points P_a, Q_a, P_c, Q_c lie on some circle ω_b and that points P_a, Q_a, P_b, Q_b lie on some circle ω_c . Can these three circles be mutually different? No, since their pairwise radical axes would in that case be the lines AB, BC, CA which are neither parallel nor concurrent (a contradiction with Proposition 1.42). Hence at least two of these three circles coincide implying that all six points lie on a single circle.

Third Proof. Direct angle-chasing is possible but in order to avoid casework caused by undetermined order of points on the triangle sides, it is convenient to use directed angles. From cyclic quadrilaterals AP_cPP_b and AQ_cQQ_b we infer

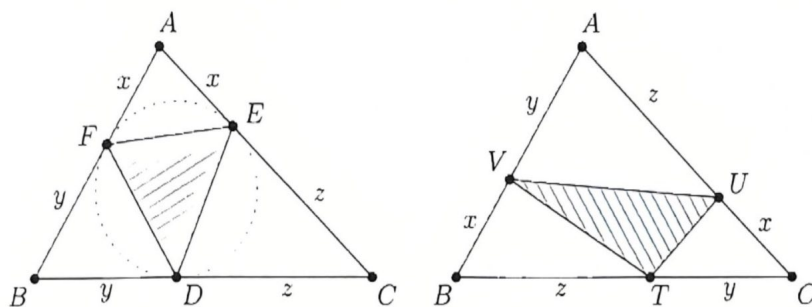
$$\begin{aligned} \angle(P_cP_b, P_bQ_b) &= \angle(P_cP_b, P_bP) + 90^\circ = \angle(P_cA, AP) + 90^\circ, \\ \angle(P_cQ_c, Q_cQ_b) &= 90^\circ + \angle(QQ_c, Q_cQ_b) = 90^\circ + \angle(QA, AQ_b), \end{aligned}$$

and thus $\angle(P_c P_b, P_b Q_b) = \angle(P_c Q_c, Q_c Q_b)$. The concyclicity of P_c, Q_c, P_b, Q_b follows and we may join any of the first two proofs.

Remark. If we take the isogonal conjugates to be H and O (orthocenter and circumcenter) the circle we obtain is the well-known *nine-point circle*. More about the nine-point circle will be discussed in the sequel to this book *107 Geometry Problems from the AwesomeMath Year-Round Program*.

29. The incircle of triangle ABC is tangent to its sides BC, CA, AB at points D, E, F , respectively. The excircles of triangle ABC are tangent to the corresponding sides of triangle ABC at points T, U, V . Show that triangles DEF and TUV have the same area.

Proof. First, we recall that $AF = BV = x$, $BD = CT = y$, and $CE = AU = z$ (see Proposition 1.15(a), (c)). Then the idea is to express both areas in terms of x, y, z and then just check an algebraic equality.



In fact, it suffices to compare the complements in triangle ABC

$$\begin{aligned}[ABC] - [DEF] &= [AFE] + [BDF] + [CED], \\ [ABC] - [TUV] &= [AVU] + [BTV] + [CUT].\end{aligned}$$

We express the areas with area formula $2K = bc \sin \angle A$ and the Extended Law of Sines in the form $\sin \angle A = a/2R = (y+z)/2R$, where R is the circumradius of triangle ABC . We obtain

$$\begin{aligned}[AFE] + [BDF] + [CED] &= \frac{1}{2} (x^2 \sin \angle A + y^2 \sin \angle B + z^2 \sin \angle C) \\ &= \frac{1}{4R} (x^2(y+z) + y^2(z+x) + z^2(x+y))\end{aligned}$$

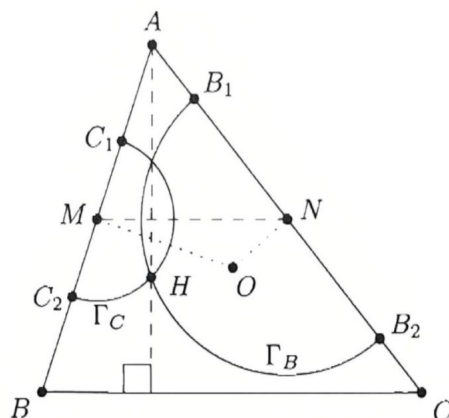
and

$$\begin{aligned}[AVU] + [BTV] + [CUT] &= \frac{1}{2} (yz \sin \angle A + zx \sin \angle B + xy \sin \angle C) \\ &= \frac{1}{4R} (yz(y+z) + zx(z+x) + xy(x+y)).\end{aligned}$$

Since the complements are indeed equal, we may conclude.

30. [IMO 2008] Let H be the orthocenter of an acute-angled triangle ABC . The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2 . Similarly, define the points B_1, B_2, C_1 , and C_2 . Prove that six points A_1, A_2, B_1, B_2, C_1 , and C_2 are concyclic.

Proof. First, we will prove that the points B_1, B_2, C_1 , and C_2 are concyclic. By the Radical Lemma (see Proposition 1.43) this is the case if and only if the radical axis of Γ_B, Γ_C passes through A . Denote the midpoints of the sides AB, AC by M, N , respectively.



The radical axis of Γ_B and Γ_C is perpendicular to MN and hence also to BC . As it passes through H , it is the A -altitude in triangle ABC . Thus, it passes through A as well and the points B_1, B_2, C_1 , and C_2 lie on one circle indeed. The center of this circle is the intersection of the perpendicular bisectors of B_1B_2 and C_1C_2 , i.e. the circumcenter O of triangle ABC .

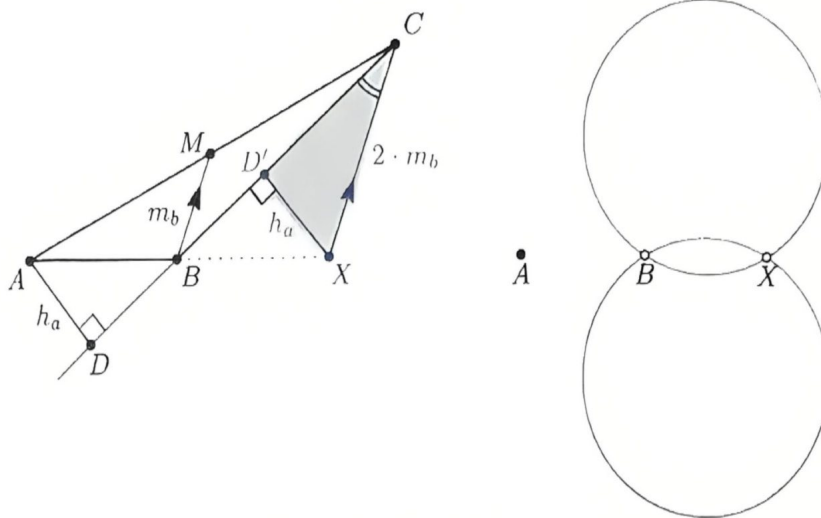
By symmetry, A_1, A_2, B_1 , and B_2 lie on a circle centered at O as well. As both these circles pass through B_1 and B_2 , they in fact coincide and the claim is proved.

31. [Moscow Math Olympiad 1969] Distinct points A, B are given in the plane. Determine the locus of points C such that in triangle ABC the length of A -altitude is the same as the length of B -median.

Solution. Suppose we found point C such that if we denote the lengths of B -median BM and A -altitude AD of triangle ABC by m_b, h_a , respectively, then $m_b = h_a$. We intend to relate AD and BM .

To this end, let X be the point such that B is the midpoint of AX . Then BM is the midline in triangle AXC so $XC = 2 \cdot m_b$. Also, if we denote the foot of perpendicular dropped from X onto BC by D' then the triangles ABD and XBD' are congruent so $XD' = h_a$. Hence in

the right triangle $CD'X$ we know $\sin \angle D'CX = \frac{1}{2}$ so $\angle D'CX = 30^\circ$ and thus $\angle BCX$ is either 30° or 150° (as it may be either congruent or supplementary to $\angle D'CX$, depending on the position of D' on BC).



On the other hand, if for some point C the measure of $\angle BCX$ is 30° or 150° then by reverse chain of thoughts we get $h_a = m_b$.

Therefore the sought locus is the union of all the points on the two circles passing through B and X whose centers form equilateral triangles with the points B and X , but the points B and X themselves.

32. [MEMO 2011, Michal Rolínek and Josef Tkadlec] Let ABC be an acute triangle with altitudes BB_0 and CC_0 . Point P is given such that the line PB is tangent to the circumcircle of triangle PAC_0 and the line PC is tangent to the circumcircle of triangle PAB_0 . Prove that AP is perpendicular to BC .

First Proof. Lines AP and BC are perpendicular if and only if

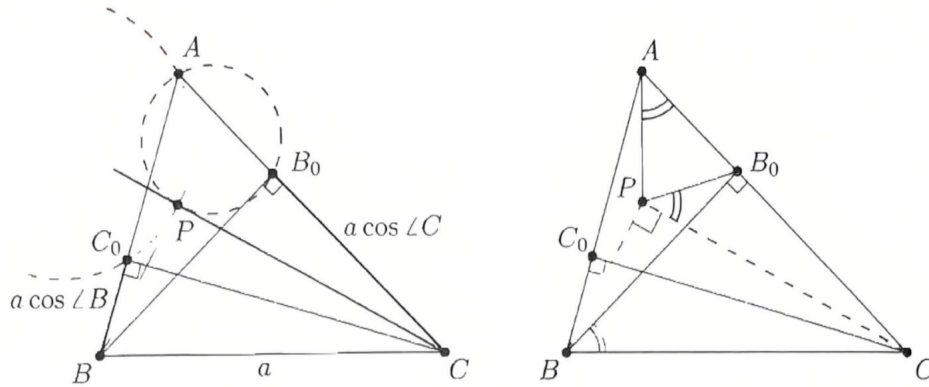
$$AB^2 + CP^2 = AC^2 + BP^2$$

(see Proposition 1.22). Using Power of a Point for B, C with respect to the circumcircles of triangles PAC_0, PAB_0 , respectively, we get rid of point P and obtain

$$\begin{aligned} AB^2 + CP^2 &= AB^2 + CA \cdot CB_0 = c^2 + b \cdot (a \cos \angle C), \\ AC^2 + BP^2 &= AC^2 + BA \cdot BC_0 = b^2 + c \cdot (a \cos \angle B), \end{aligned}$$

Finally, the Law of Cosines yields

$$ba \cos \angle C = \frac{1}{2}(a^2 + b^2 - c^2) \quad \text{and} \quad ca \cos \angle B = \frac{1}{2}(a^2 + c^2 - b^2)$$



implying that both right-hand sides of the former equalities are equal to $\frac{1}{2}(a^2 + b^2 + c^2)$.

Second Proof. Using Power of a Point as in the first proof, we obtain $BP^2 = BA \cdot BC_0 = ac \cos \angle B$ and $CP^2 = CA \cdot CB_0 = ab \cos \angle C$. Hence

$$BP^2 + CP^2 = a(c \cos \angle B + b \cos \angle C) = a^2,$$

where we have recognized $c \cos \angle B + b \cos \angle C$ as the length of BC split at the foot of the altitude from A . Hence BPC is a right triangle and BCB_0PC_0 is a cyclic pentagon. Finally, from tangency (see Proposition 1.34) and concyclicity we obtain

$$\angle PAC = \angle CPB_0 = \angle CBB_0 = 90^\circ - \angle C.$$

Hence AP is perpendicular to BC .

33. [Moscow Math Olympiad 2011] Let ABC be a triangle. Point O in its interior satisfies $\angle OBA = \angle OAC$, $\angle BAO = \angle OCB$, and $\angle BOC = 90^\circ$. Find AC/OC .

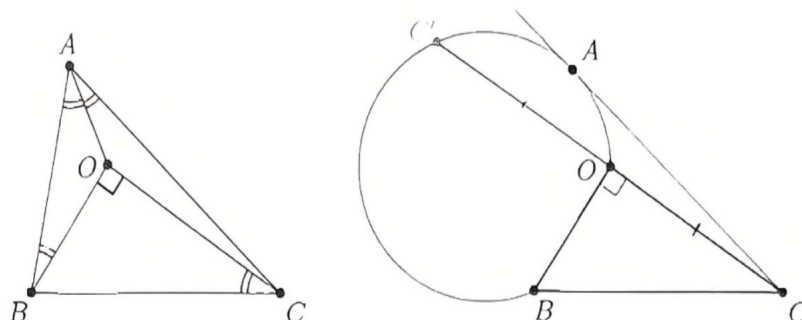
Solution. It is somewhat difficult to draw an accurate diagram for this problem since for a fixed triangle ABC there does not necessarily have to exist point O in its interior satisfying all the three requirements. Hence we start with right triangle BOC and aim to construct point A such that the remaining two equalities hold too.

Since $\angle BAO = \angle OCB$, the segment OB is visible from A and C under the same angle and consequently A has to lie on the arc $BC'O$ where C' is the reflection of C about BO . The equality $\angle OBA = \angle OAC$ then implies that CA is tangent to the circumcircle of $BOAC'$ (see Proposition 1.34), so our diagram is complete.

Now it suffices to combine Power of a Point and symmetry about BO to obtain

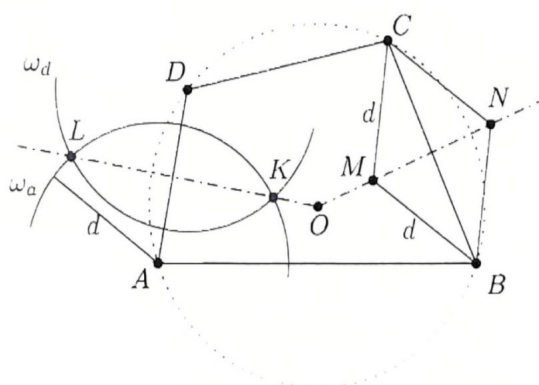
$$CA^2 = CO \cdot CC' = CO \cdot (2 \cdot CO) = 2 \cdot CO^2.$$

Therefore the answer is $AC/OC = \sqrt{2}$.



34. [Poland 2007] Let $ABCD$ be a cyclic quadrilateral ($AB \neq CD$). Quadrilaterals $AKDL$ and $CMBN$ are rhombi with equal sides. Prove that points K, L, M, N lie on a single circle.

Proof. Given a cyclic quadrilateral $ABCD$, how do we find points K and L such that $AKDL$ is a rhombus with given side length d ? Of course, we have to intersect circles ω_a, ω_d with centers A, D , respectively, and common radius d . Let us view points K, L (and likewise M, N) from this perspective instead.



As KL is the perpendicular bisector of AD , it passes through the circumcenter O of $ABCD$ and the same holds for MN , the bisector of BC . Hence it is reasonable to employ Power of a Point and try to prove $\overline{OK} \cdot \overline{OL} = \overline{OM} \cdot \overline{ON}$.

This is quickly carried out once we observe

$$\begin{aligned}\overline{OK} \cdot \overline{OL} &= p(O, \omega_a) = OA^2 - d^2, \\ \overline{OM} \cdot \overline{ON} &= p(O, \omega_b) = OB^2 - d^2,\end{aligned}$$

where we invoked the very definition of Power of a Point. Indeed, as $OA = OB$, the result follows.

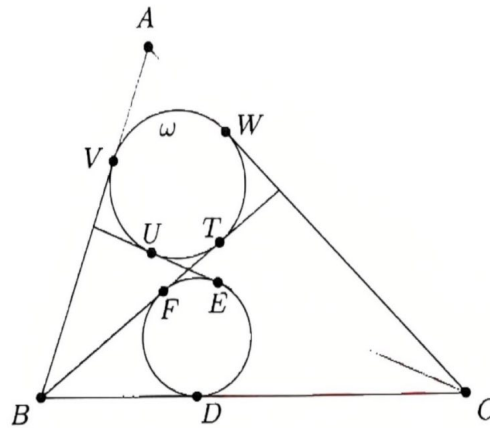
35. Let ABC be a triangle with inradius r and let ω be a circle of radius $a < r$ inscribed in angle BAC . Tangents from B and C to ω (different

from the triangle sides) intersect at point X . Show that the incircle of triangle BCX is tangent to the incircle of triangle ABC .

Proof. Denote the points of contact of the incircle of triangle BCX with its sides BC , CX , XB by D , E , F , respectively. Since both the incircles of triangle ABC and triangle BCX are tangent to BC , we aim to prove that they are tangent to it at the same point. For this it suffices to prove

$$BD - DC = \frac{1}{2}(a - b + c) - \frac{1}{2}(a + b - c) = AB - AC$$

(see Proposition 1.15).



Let BX , CX , BA , CA be tangent to ω at T , U , V , W , respectively. Then $FT = UE$ (symmetry) and by Equal Tangents,

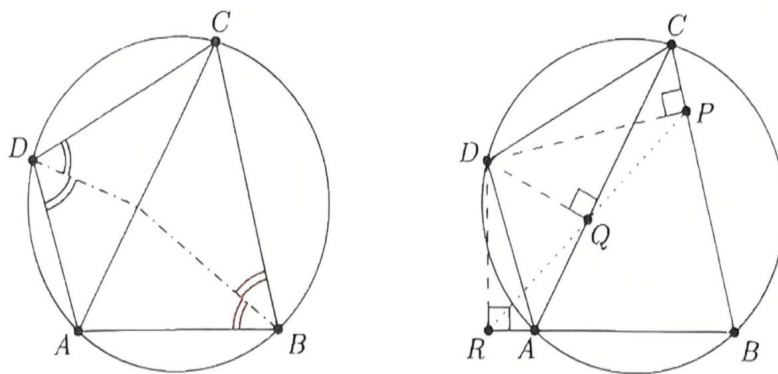
$$BD - DC = BF - EC = BT - UC = BV - WC = AB - AC,$$

which finishes the proof.

36. [IMO 2003] Let $ABCD$ be a cyclic quadrilateral. Let P , Q , R be the feet of the perpendiculars from D to the lines BC , CA , AB , respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with AC .

Proof. First, we explore the concurrence of the angle bisectors on AC as it happens if and only if both angle bisectors divide AC in the same ratio. If we use the Angle Bisector Theorem in triangles ABC and ADC , we can restate this equivalently as

$$\frac{AD}{CD} = \frac{AB}{BC}.$$



Let's now focus on the second statement. Note that points D, Q, P, C lie on a circle with diameter DC and likewise points D, Q, A, R lie on a circle with diameter AD . This enables us to find the lengths of the chords PQ and QR from the Extended Law of Sines as

$$\frac{PQ}{\sin \angle QCP} = CD, \quad \frac{QR}{\sin \angle QAR} = AD.$$

Now since $\sin \angle QCP = \sin \angle ACB$ and $\sin \angle QAR = \sin \angle BAC$, we can say that $PQ = QR$ if and only if

$$\frac{AD}{CD} = \frac{\sin \angle ACB}{\sin \angle BAC} = \frac{AB}{BC},$$

where the last equality is just the Law of Sines in triangle ABC . We have shown that both statements are equivalent to the same metric condition imposed on $ABCD$. We are done.

Remark. Note that throughout the whole proof we did not need the fact that $ABCD$ was cyclic.

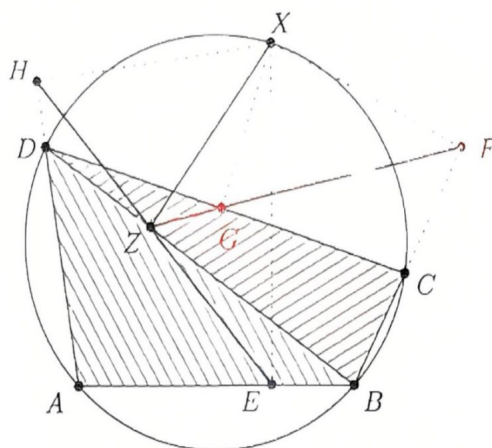
37. Let X be a point on the circumcircle of a cyclic quadrilateral $ABCD$. Denote by E, F, G , and H the projections of X onto lines AB, BC, CD, DA , respectively. Prove that

$$BE \cdot CF \cdot DG \cdot AH = AE \cdot BF \cdot CG \cdot DH.$$

Proof. We drop one more perpendicular from X , this time to the line BD , and denote its foot by Z . We recall the Simson line (see Example 1.15) and deduce that triads of points E, Z, H and Z, G, F are collinear.

In order to produce such a huge equality, we use twice Menelaus' Theorem. Once for triangle BAD and points E, Z, H and the second time for triangle BCD and points Z, G, F . We obtain

$$\frac{BZ}{DZ} \cdot \frac{DH}{AH} \cdot \frac{AE}{BE} = 1 \quad \text{and} \quad \frac{BZ}{DZ} \cdot \frac{DG}{CG} \cdot \frac{CF}{BF} = 1.$$



Expressing the ratio BZ/DZ from both equations yields

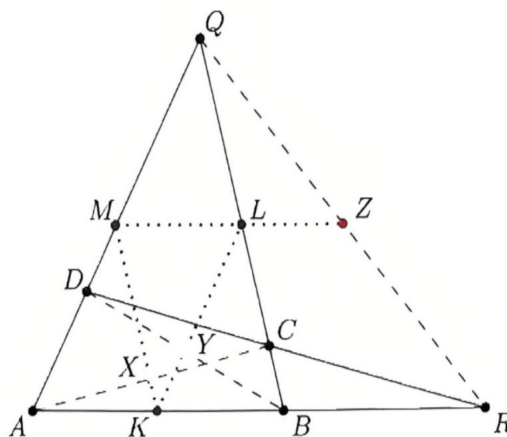
$$\frac{AH}{DH} \cdot \frac{BE}{AE} = \frac{BZ}{DZ} = \frac{CG}{DG} \cdot \frac{BF}{CF},$$

which after expanding is exactly what we wanted to prove.

38. Newton-Gauss² line.

Let $ABCD$ be a convex quadrilateral. Denote by Q the intersection of AD and BC and by R the intersection of AB and CD . Let X , Y , and Z be the midpoints of AC , BD , and QR , respectively. Prove that X , Y , and Z lie on a single line.

Proof. Proving collinearity when midpoints are involved should invoke using Menelaus' Theorem. But for which triangle? We add more midpoints to find one and as usual, we expect to produce midlines. One of the possible choices are the midpoints of the sides of triangle ABQ .



We claim that if we denote by K , L , M the midpoints of AB , BQ , and QA , then the points X , Y , and Z lie on the sides of triangle KLM

²Johann Carl Friedrich Gauss (1777–1855) was a German mathematician and physicist.